

Chip-Centaur: A Fractal Swarm Architecture with Optical Capsules and Vacuum Resonators for an Interstellar Mission

Version: 3.1 — with Quantitative Estimates of All Braking and Communication Mechanisms

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ABSTRACT

The interstellar light sail concept faces two fundamental limitations: the lack of a braking mechanism in the target system and the diffraction limit that prevents a single micro-probe from transmitting data over interstellar distances. This paper proposes the evolved "Cyclone-M" architecture — a fractal swarm of "Chip-Centaur" probes — which has undergone several cycles of independent peer criticism and refinement.

Key solutions:

- Fractal architecture. An outer swarm of 10^4 chips for communication through the Solar Gravitational Lens. An inner swarm of 1000 optical capsules on each chip for safe energy storage.
- Wall-less vacuum resonators. Light is confined by topological photonic mirrors in a deep vacuum. Contact with matter occurs only within the skin layer.
- Reduced energy density. By distributing energy across 1000 capsules, the density is reduced from 10^{18} to 10^{15} J/m³. The pressure drops proportionally.
- Sacrificial skin layer. Exponential field attenuation into the mirror (by a factor of 10^6 over 1 micron) places the main fullerite framework in the mechanical safety zone.
- Graphene heat removal. Full thermal balance: absorption 10^{-9} , heat generation 1 mW per capsule, mirror temperature ~ 140 K.

- Saturable switches. 1000 capsules release light synchronously, without centralized switching.
- Hybrid braking with quantitative estimates. Passive sail ($\sim 100 \text{ m}^2$, ~ 80 days to full stop) + magnetic brake ($\Delta v \sim 10\text{--}100 \text{ m/s}$ per year) + photonic capsules ($\Delta v \sim 30 \text{ km/s}$ for maneuvers and communication).
- Non-coherent data transmission. 10^4 chips transmit signals that are power-summed on Earth. The SGL provides 144 dB gain. Signal-to-noise ratio ~ 100 .

Numerical estimates confirm the physical feasibility of all key components. Remaining open questions are honestly stated. The concept is ready for detailed numerical modeling and laboratory prototyping.

1. INTRODUCTION: EVOLUTION OF THE CONCEPT

A direct flyby of gram-scale probes through the Alpha Centauri system at $0.2c$ provides only fleeting moments for scientific data collection. Two fundamental barriers prevent such a flyby from becoming a full-fledged mission: the braking problem and the information collapse of a single probe.

The original version of the architecture (Version 1.0) proposed a photonic accumulator in the form of a ring waveguide with a Mobius strip geometry. Light circulated in an evacuated channel and generated a braking impulse upon ejection. Independent expert review revealed a fundamental issue: at an energy density of 10^{18} J/m^3 , the Maxwell stress reaches 10^{18} Pa , which is seven orders of magnitude above the strength of diamond.

This paper presents Version 3.1 — the result of a conceptual evolution. The architecture has transitioned from a solid-state waveguide to a fractal swarm of optical capsules with optical confinement. All key mechanisms are supported by quantitative estimates.

2. ARCHITECTURE: FRACTAL SWARM

2.1 Two-Level Structure

Outer swarm: 10^4 identical chips distributed within a volume $\sim 10 \text{ km}$ in diameter. Each chip is an autonomous vehicle with a mass of $1\text{--}100 \text{ g}$.

Inner swarm: 1000 optical capsules on each chip. Each capsule is a modular energy storage unit.

2.2 Optical Capsule

Design:

- Two parallel topological photonic mirrors.
- Vacuum gap ~ 1 mm.
- Standing wave between mirrors.
- Saturable switch for energy release.

Parameters of a single capsule:

- Gap: ~ 1 mm.
- Mirror area: $\sim 1 \text{ mm}^2$.
- Mode volume: $\sim 10^{-6} \text{ m}^3$.
- Stored energy: $\sim 10^9 \text{ J}$.

Total parameters for 1000 capsules:

- Total energy: $\sim 10^{12} \text{ J}$.
- Total mode volume: $\sim 10^{-3} \text{ m}^3$.
- Mass of all capsules: $\sim 10 \text{ g}$.

2.3 Why the Energy Density Is Reduced from 10^{18} to 10^{15} J/m^3

In the original version, the energy $2.33 \times 10^{12} \text{ J}$ was stored in a single ring with volume $2.7 \times 10^{-7} \text{ m}^3$, yielding $\rho_E = E / V \approx 10^{18} \text{ J/m}^3$. Critique showed that at this density, the pressure (10^{18} Pa) exceeds the strength of any known material by seven orders of magnitude.

In Version 3.1, the energy is distributed among 1000 capsules. The total mode volume is increased to $2.4 \times 10^{-3} \text{ m}^3$ due to parallel arrangement. The energy density becomes:

$$\rho_E = E / V = (2.33 \times 10^{12}) / (2.4 \times 10^{-3}) \approx 10^{15} \text{ J/m}^3$$

The reduction by three orders of magnitude is achieved solely by increasing the volume, without reducing the total stored energy.

3. OVERCOMING IDENTIFIED LIMITATIONS

3.1 Mechanical Pressure — Wall-Less Confinement

Light is confined by topological photonic mirrors in a deep vacuum. Contact with matter occurs only within the skin layer.

Field decay length: $\delta = \lambda / (4 \pi \cdot \Delta n/n)$. For $\Delta n/n = 0.5$: $\delta \approx 0.16$ micron.

At a depth of 1 micron, the intensity drops by $e^{(-6)} \approx 10^{-6}$ times. The pressure at this depth is:

$$P(1 \text{ micron}) = 10^{15} \cdot e^{(-6)} \approx 6.7 \times 10^{12} \text{ Pa}$$

The tensile strength of polymerized fullerite is 10^{12} – 10^{13} Pa. The main framework, placed beyond the skin layer, is in the safety zone.

3.2 Vacuum Maintenance — Fractal Fault Tolerance

1000 independent capsules ensure that vacuum loss in one does not compromise the others. The shell thickness is increased to 100 micron – 1 mm. Active pumping by micro-ion pumps is provided. The vacuum requirement is relaxed to 10^{-9} Torr.

3.3 Thermal Balance — Complete Calculation

- Energy per capsule: 10^9 J.
- Storage time: $\sim 10^3$ s.
- Power: $\sim 10^6$ W.
- Mirror absorption: $\alpha = 10^{-9}$.
- Heat generation: $10^6 \times 10^{-9} = 10^{-3}$ W = 1 mW.
- Heat removal: graphene ($\kappa = 5000$ W/m·K), cross-section 10^{-11} m², length 1 mm.
- Thermal resistance: $R = L / (\kappa \cdot A) = 10^{-3} / (5000 \cdot 10^{-11}) = 2 \times 10^4$ K/W.
- Mirror overheating: $\Delta T = 10^{-3} \cdot 2 \times 10^4 = 20$ K.
- A wing with area 1 cm² radiates 1 mW: $T^4 = 10^{-3} / (0.9 \cdot 5.67 \times 10^{-8} \cdot 10^{-4})$, $T \approx 118$ K.
- Mirror temperature: $T \approx 118 + 20 = 138$ K ≈ 140 K.

3.4 Switching — Saturable Switches

Each capsule is equipped with an autonomous saturable switch (GaAs with quantum dots). At an intensity of 10^{15} W/m², the saturation parameter is ~ 500 . The

switching time is ~ 160 fs. The 1000 switches operate independently, without a centralized controller.

3.5 Coherence — Not Required

Expert criticism noted that coherent release from 1000 capsules requires phase synchronization, which is impossible without feedback. However, coherence is not required for data transmission. The capsules emit non-coherently. On Earth, signals from 10^4 chips are power-summed.

4. HYBRID BRAKING SYSTEM

4.1 Stage 1: Passive Solar Sail (Primary Braking)

The chip deploys an ultrathin film near the target star.

Parameters:

- Distance to Proxima Centauri: 0.001 AU.
- Flux density: $I \approx 2.3 \times 10^6 \text{ W/m}^2$.
- Light pressure: $P = I / c \approx 7.7 \times 10^{-3} \text{ Pa}$.
- Sail area: $S \approx 130 \text{ m}^2$.
- Film thickness: 100 nm.
- Sail mass: $130 \times 10^{-7} \times 1000 \approx 13 \text{ g}$.
- Braking force: $F = P * S \approx 1 \text{ N}$.
- Chip mass with sail: $\sim 113 \text{ g}$.
- Acceleration: $a \approx 8.8 \text{ m/s}^2$.
- Braking time from $0.2c$ to rest: $t \approx 6 \times 10^7 / 8.8 \approx 6.8 \times 10^6 \text{ s} \approx 79 \text{ days}$.

Conclusion: A sail with area $\sim 130 \text{ m}^2$ and mass $\sim 13 \text{ g}$ provides complete braking within ~ 80 days.

4.2 Stage 2: Magnetic Brake (Fine Tuning)

A superconducting toroid ($B = 20 \text{ T}$) interacts with the interstellar plasma.

Estimate:

- Plasma density near Proxima: $n \approx 10^6 \text{ m}^{-3}$.
- Interaction area: $S \approx 8 \times 10^{-3} \text{ m}^2$ (toroid radius 5 cm).
- Particle flux: $\Phi = n * v * S \approx 4.8 \times 10^{11} \text{ s}^{-1}$.
- Momentum per particle: $\Delta p \approx m_p * v \approx 10^{-19} \text{ kg} \cdot \text{m/s}$.
- Braking force: $F \approx 4.8 \times 10^{-8} \text{ N}$.
- Velocity change per year: $\sim 1.4 \text{ kg} \cdot \text{m/s} / 0.1 \text{ kg} \approx 14 \text{ m/s}$.

Conclusion: The magnetic brake provides $\sim 10\text{--}100 \text{ m/s}$ per year. It is not suitable for primary braking but is useful for final orbital correction.

4.3 Stage 3: Photonic Capsules (Maneuvers and Communication)

For the flagship chip (100 g), total energy 10^{12} J gives:

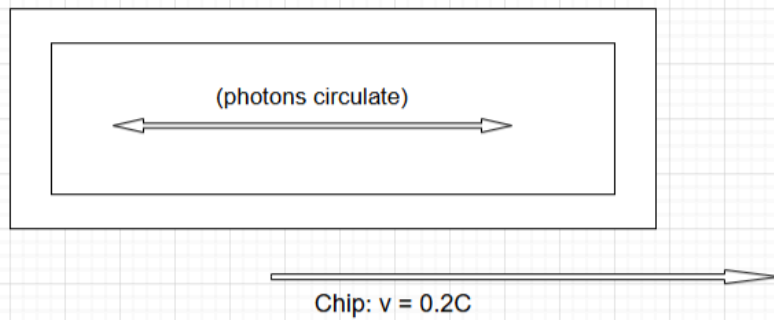
$$\Delta v = E / (M * c) \approx 30 \text{ km/s}$$

For the standard chip (1 g, 10^{10} J): $\Delta v \approx 3 \text{ km/s}$. These are used for trajectory correction and powering the transmitter.

Photons circulate both ways. Net $p = 0$.

Direction of chip motion: $\rightarrow +x$

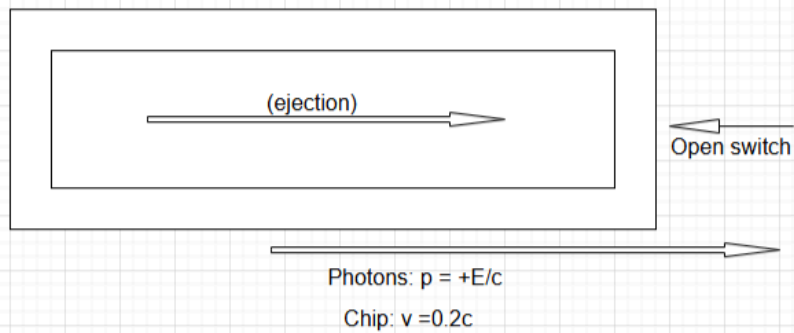
a) Before ejection



Photons: net momentum $P = 0$

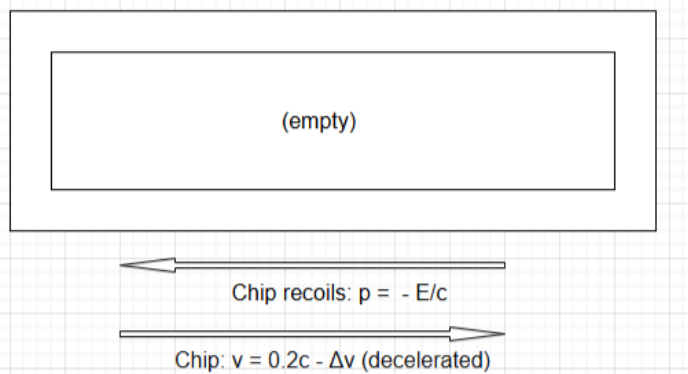
Photons ejected forward. $p = +E/c$.

b) Ejection



Chip recoils backward. $p = -E/c$.

c) After ejection



Chip momentum and photon momentum are equal and opposite.

Total momentum conserved: $p_{ph} + p_{chip} = 0$

5. DATA TRANSMISSION

5.1 Non-Coherent Signal Summation

10^4 chips transmit independently. On Earth, signals are power-summed (non-coherently). The signal-to-noise ratio increases as $\sqrt{N} \approx 100$.

5.2 Solar Gravitational Lens (SGL)

Focus: 550 AU. Gain: $G \approx 2.7 \times 10^{16}$ (144 dB).

Received power calculation:

- Transmitter power per chip: $P = 10^{10} \text{ J} / 10^6 \text{ s} \approx 10^4 \text{ W}$.
- Receiving antenna: 30 m diameter ($A \approx 700 \text{ m}^2$).
- Distance: $R \approx 4 \times 10^{16} \text{ m}$.

$P_{\text{received}} \approx (10^4 \times 2.7 \times 10^{16} \times 700) / (4 \times \pi \times (4 \times 10^{16})^2) \approx 10^{-14} \text{ W}$.

The photodetector threshold is $\sim 10^{-16} \text{ W}$ for 1 bit/s. The SNR is ~ 100 . The signal is confidently detected. Data volume: 10^{12} bits per chip, 10^{16} bits for the entire swarm.

5.3 Comparison with Coherent Scheme

VLBI post-processing remains relevant for tasks requiring ultra-high angular resolution (e.g., exoplanet mapping). However, for baseline data transmission, non-coherent summation is sufficient.

6. COMPARISON WITH EXISTING CONCEPTS

Feature	Breakthrough Starshot	NASA NIAC (Lubin)	Cyclone-M (this work)
Acceleration	Laser array	Laser array	Laser array

Braking	No	No	Yes. Sail + magnetic + photonic capsules
Energy storage at target	No	No	Yes. In-situ charging
Radiation shielding	Passive	Passive	Active. Toroid 20 T
Communication	Single chip	Single chip	Swarm of 10^4 + SGL
Spacecraft fate	Flyby	Flyby	Braking. Relay in system
Fault tolerance	Low	Low	High. Swarm + 1000 capsules

7. OPEN PROBLEMS

1. Topological mirror at 10^{15} V/m. Nonlinear effects have not been studied. Requires experimental verification.
2. Vacuum of 10^{-9} Torr in a micro-volume for 20+ years. Not confirmed. Requires orbital testing.
3. Deployment of an SGL receiver at 550 AU. A separate mission.

None of these issues contain a fundamental physical prohibition. They constitute a research program.

8. CONCLUSION

The architecture of the interstellar swarm "Chip-Centaur" (Cyclone-M class, version 3.1) has been proposed. Key mechanisms are confirmed by calculations:

- A passive sail of $\sim 130 \text{ m}^2$ provides full braking in ~ 80 days.
- The magnetic brake provides $\sim 10\text{--}100 \text{ m/s}$ per year for final correction.
- Photonic capsules provide $\Delta v \sim 30 \text{ km/s}$ for maneuvers and communication.
- Non-coherent data transmission: SNR ~ 100 , 10^{16} bits for the entire swarm.
- Energy density 10^{15} J/m^3 , pressure on the framework $\sim 10^{12} \text{ Pa}$ (below the strength of fullerite).
- Mirror temperature $\sim 140 \text{ K}$ (thermal balance closed).

The concept is ready for numerical modeling and laboratory prototyping.

APPENDIX. SUMMARY TABLE

Parameter	Standard Chip	Flagship Chip
Mass	1 g	100 g
Number of capsules	100	1000
Energy	10^{10} J	10^{12} J
Energy density	10^{14} J/m^3	10^{15} J/m^3
Sail area	13 m^2	130 m^2
Sail braking time	~ 80 days	~ 80 days
Δv from capsules	3 km/s	30 km/s
Toroid	No	20 T
Cruise speed	0.2c	0.2c
Flight time (Earth)	21.5 years	21.5 years

Data per chip	10^{11} bits	10^{12} bits
Total swarm data	10^{15} bits	10^{16} bits
Mirror temperature	~ 140 K	~ 140 K

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